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Research paper

Modelling the impact of renewable energy investment on global carbon dioxide emissions

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ABSTRACT

The adversities of climate change facing the world highlight the importance of reducing the rise of global carbon emissions (CO_2E), which is a top priority for policymakers and researchers. Renewable energy is hailed as a remedy to the rise in global CO_2E. However, there are persistently low levels of global renewable energy investment (REI). This study, therefore, examines the linear and non-linear effect of REI (measured by solar, wind, small hydropower, geothermal, biomass, marine, and biofuel energy investment) on CO_2E using a global dataset from 2004Q1 to 2018Q1. The study also examines the economic growth mechanism through which REI influences CO_2E. Using the Ordinarily Least Squares model, the findings show that all the measures of REI significantly increase CO_2E from the linear analysis. However, the non-linear analysis showed that solar and wind energy investment have a statistically significant inverted U-shaped relationship with CO_2E while small hydropower and biofuel energy investment have a statistically significant U-shaped relationship with CO_2E. Thus, solar and wind energy investment reduce CO_2E only after attaining a threshold of investment with the threshold value of solar and wind energy investment being 28.125 (representing about $1.64 trillion) and 26.308 (representing about $266.3 billion), respectively. Also, the threshold value of small hydropower and biofuel energy investment beyond which they increase CO_2E are 20.071 (representing about $520.9 million) and 22.8 (representing about $7.978 billion), respectively. Finally, solar, wind and geothermal energy investment indirectly reduce CO_2E by increasing green economic growth while the reverse exists for biofuel energy investment. Solar and wind energy investment is, therefore, recommended for attaining carbon neutrality targets. Policymakers should incentivize production tax credits, investment tax credits, feed-in-tariffs, subsidies, affordable loans, and research and development to drive solar and wind energy investment.

1. Introduction

Global carbon emissions (CO_2E) have been increasing since the industrial revolution reaching a record of 34 million metric tons in 2018 from as low as 21 million metric tons in 1990 (see Fig. 1). The problem with this surge in global CO_2E is climate change with its attendants impacts such as droughts, floods, wildfires, storms, rising sea level, and the rising global temperatures. In 2017, climate change resulted in the globe losing $320 billion and it is projected that over 100 million people will fall into poverty by 2030 with about 143 million people being climate migrants by 2050 due to climate change (World Bank Group, 2019). To this end, lowering CO_2E, the defining factor of climate change, is widely acclaimed among policymakers (Appiah-Otoo et al., 2022a; Appiah-Otoo and Acheampong, 2021; Qin and Gong, 2022). For instance, the Sustainable Development Goal (SDG) 13 establishes that countries should take decisive action to reduce climate change and its repercussions by 2030. Also, the 2015 Paris Agreement stipulates countries keep global temperatures from rising over 2 degrees Celsius.

Achieving these tasks requires the revolution of the worldwide socioeconomic system from fossil fuel consumption to renewable energy (RE) sources such as solar, wind, and hydropower which are usually clean and easily accessible everywhere (Appiah-Otoo et al., 2023a; b). RE has the potential to also promote energy independence, energy security, and sustainable economic growth (Appiah-Otoo, 2021). Accordingly, the recent period has witnessed a surge in renewable energy investment (REI) from as low as $45.3 billion in 2004 to an astounding $288.3 billion in 2018 representing a change of $243 billion in REI (see Fig. 2). This rapid investment has also translated into a surge

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in RE generation (REG) progressing from as low as 3144 terawatt-hours (TWh) in 2004–6672 TWh in 2018 (see Fig. 3). Theoretically, expanding REI can decrease CO_2E by facilitating a shift towards cleaner energy sources and reducing energy intensity (Yang et al., 2022). However, the empirical studies on the environmental impact of REI remain inconclusive. For instance, while Yang et al. (2022) found that REI raises CO_2E, Zhang et al. (2021) found the contrary. This calls for a new study to address these conflicts. To this end, this study is motivated to examine: (i) the impact of seven different measures of REI (measured by solar, wind, small hydropower, geothermal, biomass, marine, and biofuels energy investment) on CO_2E using a global dataset, (ii) the economic growth mechanism via which REI influences CO_2E, (iii) the non-linear relationship between REI and CO_2E, (iv) the threshold value beyond which REI could either reduce or increase CO_2E, (v) the optimal REI amount needed before REI could either reduce or increase CO_2E and (vi) the co-movement between REI and CO_2E.

The contributions of this study are five-fold. First, this study contributes to the literature that examines the environmental impact of REI by using global datasets. Previous research on this topic has focused countrywide basis such as China (Zhang et al., 2021), India (Mahesh and Shoba Jasmin, 2013), and other leading REI countries (He et al., 2023; Yang et al., 2022). However, with climate change being a global phenomenon, this calls for a global dataset to make globally informed decisions. Also, this study employed seven different measures of REI (solar, wind, small hydropower, geothermal, biomass, marine, and biofuels energy investment) for the empirical analysis. Previous studies have mostly focused on aggregate REI. However, different measures of REI have varied impacts on CO_2E, thus the need to take into consideration the seven different measures of REI. Moreover, this study explored the non-linear relationship between REI and CO_2E and also presented the threshold value and the optimal REI amount needed beyond which REI could either reduce or increase CO_2E. This adds to the various REI projections made by international organizations. For instance, the International Renewable Energy Agency (IRENA) projects that the world will require over $1.7 trillion in REI by 2030 to reach the carbon neutrality target (IRENA, 2017). However, their projection was based on aggregate REI rendering their projections difficult for policy purposes. Furthermore, this study uses the wavelet coherence (WC) analysis to model the interdependence between REI and CO_2E. This is not available in the REI and CO_2E literature. Finally, the findings of this study will guide in developing prudent REI and CO_2E policies to help achieve the carbon neutrality target.

The remaining sections of the study comprise the literature review and hypothesis construction in Section 2. Section 3 shows the methods and data used in this study while Section 4 displays the findings and discussion of the study. The final section (Section 5) concludes the study and offers policy insights.

2. Review of literature and hypotheses

2.1. Literature review

2.1.1. Renewable energy investment and CO_2E

Theoretically, REI can have both positive and negative effects on CO_2E. First, increasing the scale of REI can stimulate economic growth and lead to higher energy consumption and CO_2E (Yang et al., 2022). Conversely, expanding REI can also decrease CO_2E by facilitating a shift toward cleaner energy sources and also by reducing energy intensity (Zhang et al., 2021). Against these conflicting theories, the empirical literature on REI and CO_2E is also varied. Mahesh and Shoba Jasmin (2013) conducted an extensive review of the possibility of REI in reducing CO_2E in India and concluded that REI reduces CO_2E. Zhang et al. (2021) studied the linear and non-linear impact of REI on CO_2E in China and found that REI reduces CO_2E from the linear model while the non-linear impact remained mixed (REI increases CO_2E in the short-run, reduces CO_2E in the medium-run and increases CO_2E in the long-run). However, this study only concentrated on aggregate REI while ignoring the optimal REI and co-movement between REI and CO_2E. Yang et al. (2022) also found that REI increases CO_2E for 13 countries with high REI investment. Their study also showed that REI reduces CO_2E by reducing energy intensity; however, it increases CO_2E through the economic growth mechanism. This study was limited to only 13 countries and four sub-measures of REI (solar, wind, hydropower, and biomass energy investment). Also, this study ignored the non-linear, optimal REI, and co-movement between REI and CO_2E. He et al. (2023) study also focused on leading REI countries and found that REI reduces CO_2E and indirectly via export. However, this study also focused on fewer countries and also ignored the non-linear, the optimal REI, and the co-movement between REI and CO_2E. Accordingly, this study is an extension of the prior REI and CO_2E studies by using global datasets.

2.1.2. Population, economic growth, trade openness, and CO_2E

Population, economic growth, and trade openness are also drivers of CO_2E. The environment is negatively affected by the increasing energy demand caused by population growth (Shi, 2003). Furthermore, the expansion of the population can lead to alterations in land use, including the utilization of wood fuel and deforestation (Shi, 2003). Conversely, the rapid growth of the population stimulates technological progress,

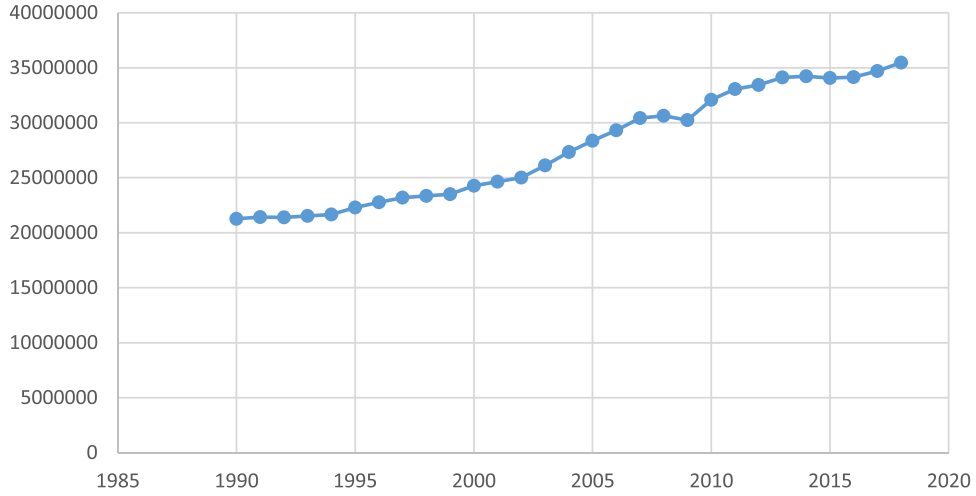


Fig. 1. Trend in global carbon emissions in metric tons.
(Source of data: World Development Indicators (WDI)).

resulting in environmental benefits (Adams et al., 2020). Numerous studies have empirically demonstrated that population growth contributes to an increase in CO_2E (Adams et al., 2020; Bai et al., 2019; Yu et al., 2023).

According to the Environmental Kuznets Curve (EKC) theory, an initial increase in income leads to higher CO_2E, but once a certain threshold is reached, CO_2E starts to decline (Grossman and Krueger, 1991). Several studies (Raihan et al., 2023; Ridzuan et al., 2020; Nie et al., 2019 and Shabani and Shahnazi, 2019) support the notion that economic growth contributes to increased CO_2E. However, other studies, such as those by (Appiah-Otoo and Acheampong, 2021) and (Alaganthiran and Anaba, 2022), have found the opposite trend.

The impact of trade openness on the environment remains inconclusive. Trade openness is a significant factor in economic growth (Appiah-Otoo and Song, 2022), which can lead to higher consumption of polluting energy sources and subsequently higher CO_2E. On the other hand, trade openness can also drive the adoption of cleaner energy technologies, thus improving environmental quality, as has been highlighted (Shahbaz et al., 2013, 2019). Empirical evidence from studies (Opoku-Mensah et al., 2021; Al-Mulali et al., 2015; Dauda et al., 2021; Kasman et al., 2015) supports the argument that trade openness contributes to environmental degradation. However, other studies (Shahbaz et al., 2013, 2019) suggest the opposite relationship.

Table 1 is a summary of some of the selected studies on the impact of REI, economic growth, population size, and trade openness on CO_2E.

2.2. Hypotheses

RE technologies are relatively new and are characterized by low levels of adoption and efficiency (Appiah-Otoo, 2021; Yang et al., 2022). There is also insufficient knowledge on how to effectively harness these technologies and are mostly accompanied by high installation and maintenance costs, which may discourage investment due to uncertain returns (Appiah-Otoo, 2021; Yang et al., 2022). Also, the production, processing, distributing, and fixing of RE technologies require the use of dirty energy consumption, resulting in more CO_2E being released into the atmosphere (Yang et al., 2022). For instance, Yang et al. (2022) noted that REI increases CO_2E. This leads to our first hypothesis.

H1. . REI increases CO_2E.

Increasing REI is also associated with rising economic growth. For instance, Yang et al. (2022) found that REI increases economic growth while economic growth is also a key driver of CO_2E (Nie et al., 2019; Ridzuan et al., 2020; Sharma, 2011). Thus, we hypothesize.

H2. . REI indirectly increases CO_2E through economic growth.

REI will initially increase CO_2E given our prior stated hypotheses; however, after a certain stage of development, REI might reduce CO_2E via their high levels of adoption by improving efficiency, green economic growth, and reducing energy intensity. Therefore, we hypothesize:

H3. . There is an inverted U-shaped relationship between REI and CO_2E.

3. Data and Methodology

3.1. Data

This study focuses on global data (world aggregates) ranging from 2004Q1 to 2018Q1. The data and the period were chosen based on the accessibility of the data. We used the variables CO_2E, population size, GDP per capita (representing economic growth), trade openness, and REI (represented by solar, wind, small hydropower, geothermal, biomass, marine, and biofuel energy investment) for the empirical analysis. The REI data was obtained from Our World in Data[1] while the data on the rest of the variables were sourced from the World Bank's database (WDI, 2023). The parameters are log-transformed to produce elasticities.

Table 2 displays the variables, code, definition, unit of measurement, justification and data source.

Table 3 displays the variables' descriptive statistics. The mean of CO_2E, population, economic growth, and trade openness is 17.250%, 22.677%, 9.163%, and 4.051%, respectively. Regarding the REI measures, solar and wind energy investment have the highest averages of 25.077% and 25.037%, respectively. This shows the dominance of solar and wind energy in the total RE mix. Followed closely are biomass energy investment (23.256%), biofuel energy investment (22.736%), small hydropower energy investment (22.414%), and geothermal energy investment (21.446%) while marine energy investment has the lowest average (19.298%). The standard deviation of the variables is below the mean suggesting that the variables are not volatile.

The Augmented Dickey-Fuller (ADF) (Dickey and Fuller, 1979) unit root test was employed to examine the unit root properties of the variables. As shown in Table 3, the variables were non-stationary at the log level ($I(0)$); however, after the first difference ($I(1)$), all the variables were stationary. Thus, the variables are integrated at the first difference ($I(1)$).

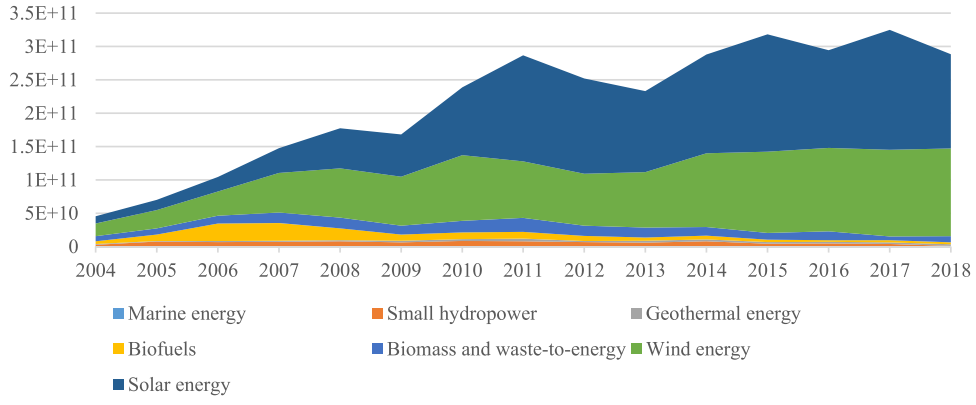


Fig. 2. Trend of renewable energy investment.
(Source of data: Our World in Data).

[1] https://ourworldindata.org/grapher/investment-in-renewable-energy-by-technology

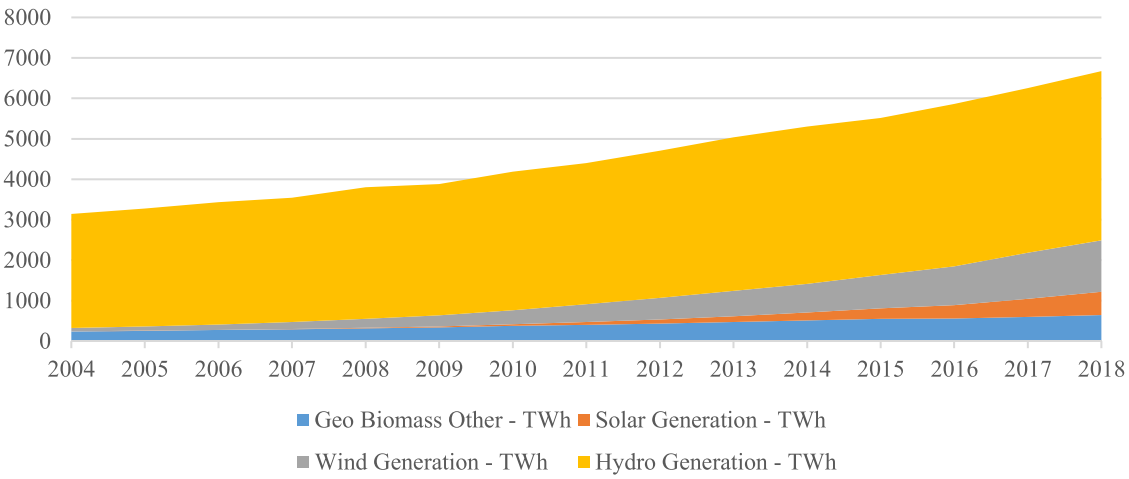


Fig. 3. Trend of renewable energy generation.
(Source of data: Our World in Data).

3.2. Empirical model

The IPAT model was developed by Ehrlich and Holdren (1971) to examine the environmental consequences of anthropogenic activities (Ntiamoah et al., 2023). This technique laid down the mathematical basis for empirically assessing the drivers of environmental pollution. However, this technique was criticized for failing to account for non-monotonic environmental parameters (York et al., 2003). In 1997, Dietz and Rosa (1997) extended the IPAT technique to STIRPAT (Stochastic impact through a regression on population, affluence, and technology) which is a more robust technique for examining the connections between human activities and the environment (Dietz and Rosa, 1997; York et al., 2003).

The STIPAT model is expressed as Eq. (1):

$$I_t = aP_t^b A_t^c T_t^d e_t \tag{1}$$

Expressing (1) in logarithm form produces (2):

$$\ln I_t = a + b\ln P_t + c\ln A_t + d\ln T_t + e_t \tag{2}$$

Where *I*represents the environment, *P* signifies the population size, *A* indicates economic growth, *T* is technology, *t* denotes time,*e* is the error term. *T*indicates factors that have an impact on the environment besides population and wealth. As a result, we modify the model to include, trade openness, and REI.

The modified model is expressed as the following:

$$\ln co2_t = \alpha_o + \alpha_1 \ln pop_t + \alpha_2 \ln gdp_t + \alpha_3 \ln trade_t + \alpha_4 \ln rei_t \tag{3}$$

To ascertain the economic growth channel via which REI influences CO_2E, we compute Eq. (4).

$$\ln co2_t = \alpha_o + \alpha_1 \ln pop_t + \alpha_2 \ln gdp_t + \alpha_3 \ln trade_t + \alpha_4 \ln rei_t + \alpha_5 \ln rei_t \times \ln gdp_t \tag{4}$$

Finally, to estimate the non-linear impact of REI on CO_2E, we compute Eq. (5).

$$\ln co2_t = \alpha_o + \alpha_1 \ln pop_t + \alpha_2 \ln gdp_t + \alpha_3 \ln trade_t + \alpha_4 \ln rei_t + \alpha_5 \ln rei^2{}_t \tag{5}$$

where ln *co2* is CO_2E, ln *pop* is the population, ln *gdp* is economic growth, ln *trade* is trade openness, ln *rei* is renewable energy investment (measured by solar (ln *solar*), wind (ln *wind*), small hydropower (ln *hydro*), geothermal (ln *geo*), biomass (ln *biomass*), marine (ln *mane*), and biofuel energy investment (ln *biofuel*)).

3.3. Econometric model

We adopt the OLS estimator to estimate the prior models because of our small sample size. This estimator provides an unbiased estimate and has the lowest variance regarding all linear unbiased estimation techniques. The study also employs the fully modified OLS (FMOLS) model, which addresses endogeneity issues and is particularly suitable for datasets with a fewer number of observations (Appiah-Otoo et al., 2022b; Appiah-Otoo and Song, 2022) for robustness analysis.

The FMOLS model is presented as Eq. 6:

$$\widehat{\beta}_{FMOLS} = \left(\sum_{t=1}^{T} x_t x'_t\right)^{-1}\left(\sum_{t=1}^{T}\left(x_t y_t^{+} - T\left[\widehat{\lambda}_{12}^{+'}0\right]\right)\right) \tag{6}$$

where

$$y_t^{+} = \widehat{\lambda}_{12} - \widehat{\omega}_{12}\Omega_{22}^{-1}\widehat{\nu}_2 \tag{7}$$

$\widehat{\lambda}_{12}^{+}$ is characterized as:

$$\widehat{\lambda}_{12}^{+} \equiv \widehat{\lambda}_{12} - \widehat{\omega}_{12}\Omega_{22}^{-1}\widehat{\Lambda}_{22} \tag{8}$$

Ω and Λ indicate covariance matrices. $\widehat{\nu}_t$ indicates residuals. ω implies a lower triangular matrix. y_t and x_t imply dependent and independent variables. $y_t^{+}$and $\widehat{\lambda}_{12}^{+}$ solves endogeneity and serial correlation challenges.

3.4. Wavelet model

This study uses the WC approach to examine the co-movement between the REI and CO_2E. This approach disaggregates the time series data into a bi-dimensional time-frequency domain (Goupillaud et al., 1984). The model is specified in Eq. 9.

$$\psi(t) = \pi^{\frac{-1}{4}} e^{-i\omega_0 t} e^{\frac{-1}{2}t^2}, p(t), t = 1, 2, 3, ., T \tag{9}$$

Where ψrepresents the mother wavelet function.

The wavelet transformed with location and frequency is presented in Eq. 10.

$$\psi_{k,f}(t) = \frac{1}{\sqrt{h}}\mu\left(t - k\Big/f\right), k, f \in R, f \neq 0. \tag{10}$$

Eq. 11 shows the continuous wavelet function with $p(t)$

$$W_p\left(k,f\right) = \int_{-\infty}^{\infty} p(t)\frac{1}{\sqrt{f}}\psi\left(\frac{\bar{t}-\bar{k}}{f}\right)dt, \tag{11}$$

Table 1
Summary of selected literature.

Author	Area of Study	Period	Methodology	Findings
(Mahesh and Shoba Jasmin, 2013)	India	-	Review	REI reduces CO_2E
(Zhang et al., 2021)	China	2004–2019	Non-parametric additive regression	REI reduces CO_2E in the linear model
(Yang et al., 2022)	13 countries	2010–2019	OLS FE Sys-GMM	REI increases CO_2E
(Bai et al., 2019)	China	2006–2013	2SLS	Population increases CO_2E
(Yu et al., 2023)	Jiangsu, China	2005–2019	Ridge regression	Population increases household CO_2E
(Adams et al., 2020)	19 SSA countries	1980–2011	IV-GMM FE	Population increases CO_2E
(Raihan et al., 2023)	Egypt	1990–2019	DOLS	Economic growth increases CO_2E
(Añón Higón et al., 2017)	116 countries	1995–2010	POLS FE IV-FE	Inverted U-shaped relationship
(Nie et al., 2019)	China	1995–2014	PSTR	Economic growth increases CO_2E
(Ridzuan et al., 2020)	Malaysia	1978–2016	ARDL FMOLS DOLS	Economic growth increases CO_2E
(Shabani and Shahnazi, 2019)	Iran	2002–2013	DOLS	Economic growth increases sectoral CO_2E
(Appiah-Otoo and Acheampong, 2021)	5 countries	2000–2016	IV-GMM ARDL	Economic growth reduces CO_2E
(R.R. R. Alaganthiran and Anaba, 2022)	20 SSA	2000–2020	POLS FE R-E	Economic growth reduces CO_2E
(Opoku-Mensah et al., 2021)	25 countries	2000–2015	ARDL	Trade openness increases CO_2E
(Al-Mulali et al., 2015)	Vietnam	1981–2011	ARDL	Trade openness increases CO_2E
(Dauda et al., 2021)	9 countries	1990–2016	FE POLS Sys-GMM	Trade openness increases CO_2E in South Africa and Mozambique
(Kasman and Duman, 2015)	1992–2010	15 countries	FMOLS	Trade openness increases CO_2E
(Shahbaz et al., 2013)	South Africa	1965–2008	ARDL	Trade openness reduces CO_2E
(Shahbaz et al., 2019)	USA	1965–2016	ARDL	Trade openness reduces CO_2E

Note: ARDL is autoregressive distributed lag, OLS is ordinary least squares, POLS is pooled OLS, FMOLS is fully modified OLS, DOLS is dynamic OLS, FE is fixed effects, R-E is random effects, IV-FE is instrumental variable FE, sys-GMM is a two-step system generalized method of moments, PSTR is panel smooth regression, SSA is sub-Saharan Africa, USA is The United States of America.

Where k and f are time and frequency domain determinants.

Eq. 12 shows the transformed model with the addition of ψ's coefficient to Eq. 11.

$$p(t) = \frac{1}{c_\psi}\int_0^\infty \left[\int_{-\infty}^{\infty} |W_p(a,b)|^2 da\right]\frac{db}{b^2}, \tag{12}$$

The time series' cross wavelet transform $W_p(k,f)$ and $W_q(k,f)$ of the variables $p(t)$ and $q(t)$ are shown as Eq. 13.

$$W_{pq}(k,f) = W_p(k,f)\overline{W_q(k,f)} \tag{13}$$

The squared wavelet function is shown in Eq. (14).

$$R^2\left(k,f\right) = \frac{\left|c\left(f^{-1}W_{pq}(k,f)\right)\right|^2}{c\left(f^{-1}\left|W_p(k,f)\right|^2\right)c\left(f^{-1}\left|W_q(k,f)\right|^2\right)}, \tag{14}$$

Where c denotes the smoothing process and time. There is a correlation among the variables when $R^2(k,f)$ approaches 1 and it is shown in red colour. However, when $R^2(k,f)$ approaches 0, it means there is no correlation and this is represented by the blue colour.

Eq. (15) shows the phase difference of the WC function.

$$\varphi_{pq}\left(k,f\right) = \tan^{-1}\left(\frac{L\left\{C\left(f^{-1}W_{pq}(k,f)\right\}\right.}{O\left\{C\left(f^{-1}W_{pq}(k,f)\right)\right\}}\right), \tag{15}$$

Where L represents an imaginary operator while O signifies a real part operator.

In WC analysis, if the arrows in the thick black area point towards the right (or left), it signifies a strong positive (or negative) correlation between the variables. Arrows pointing toward the left-up or right-down suggest that the first variable is causing the second one, while arrows pointing toward the left-down or right-up indicate that the second variable is causing the first one.

4. Results and Discussions

4.1. Direct effect analysis

Table 4 shows the direct effect results. The results show that all the measures of REI have a statistically significant positive impact on CO_2E at the 1% level. Specifically, a 1% increase in solar, wind, small hydropower, geothermal, biomass, marine, and biofuels energy investment increases CO_2E by 0.021%, 0.027%, 0.016%, 0.026%, 0.029%, 0.015%, and 0.015%, respectively. Thus, REI increases global CO_2E. This could mean that REI increases economic growth thereby increasing dirty energy consumption and CO_2E. Also, the production, distribution, and recycling of RE technologies are associated with increasing CO_2E confirming our H1. This also confirms the findings of Yang et al. (2022). However, this contradicts other findings (He et al., 2023, 2013).

Population size has a significant positive impact on CO_2E in all the models. Thus, an increase in population size leads to increased energy consumption thereby endangering the environment (Shi, 2003). This confirms the findings of Appiah-Otoo and Acheampong (2021), Bai et al. (2019), and Adams et al. (2020). Economic growth improves environmental quality in all the models. This could imply that the world has exceeded the economic growth threshold beyond which a rapid rise in economic growth increases CO_2E. This affirms the findings of Appiah-Otoo and Acheampong (2021) and Alaganthiran and Anaba (2022). Trade openness is a significant driver of CO_2E in all the models. Thus, an increase in trade drives dirty energy consumption usually via the increase in economic activities thereby increasing CO_2E. This confirms the findings of Opoku-Mensah et al. (2021), Al-Mulali et al. (2015), Dauda et al. (2021), and Kasman and Duman (2015).

Table 2
Variables definition.

Variables	Code	Definition	Unit of Measurement	Justification	Source
CO_2E	lnco2	Carbon emissions from fossil fuels combustion and the production of cement.	kiloton	(Shahbaz et al., 2019, 2013)	WDI*
Population	lnpop	Total population of a country	Total	(Appiah-Otoo et al., 2022a; Appiah-Otoo and Acheampong, 2021)	WDI*
Economic growth	lngdp	GDP per capita denotes economic growth.	constant 2015 US$ billion	(Nie et al., 2019; Raihan et al., 2023; Ridzuan et al., 2020; Shabani and Shahnazi, 2019)	WDI*
Trade openness	lntrade	Total of export and import divided by GDP	Percent	(Al-Mulali et al., 2015; Dauda et al., 2021; Kasman and Duman, 2015; Opoku-Mensah et al., 2021)	WDI*
Solar energy investment	lnsolar	Total investment in solar energy technologies	US$ billion	Yang et al. (2022)	Our World in Data
Wind energy investment	lnwind	Total investment in wind energy technologies	US$ billion	Yang et al. (2022)	Our World in Data
Small hydropower energy investment	lnhydro	Total investment in small hydropower energy technologies	US$ billion	Yang et al. (2022)	Our World in Data
Geothermal energy investment	lngeo	Total investment in geothermal energy technologies	US$ billion	(Palomo-Torrejón et al., 2021)	Our World in Data
Biomass energy investment	lnbiomass	Total investment in biomass and waste-to-energy technologies	US$ billion	Yang et al. (2022)	Our World in Data
Marine energy investment	lnmarine	Total investment in marine energy technologies	US$ billion	(Chen et al., 2022)	Our World in Data
Biofuel energy investment	lnbiofuel	Total investment in biofuels energy technologies	US$ billion	(Naqvi et al., 2023)	Our World in Data

* World Development Indicators (WDI)

Table 3
Descriptive statistics.

Variables	Obs.	Mean	Std. Dev.	Min	Max	ADF *I* (0)	ADF *I* (1)
lnco2	57	17.250	0.075	17.090	17.350	1.680	-1.908*
lnpop	57	22.677	0.050	22.591	22.760	-0.796	-2.069**
lngdp	57	9.163	0.067	9.036	9.285	2.126	-1.718*
lntrade	57	4.051	0.036	3.960	4.110	0.122	-3.093***
lnsolar	57	25.077	0.875	23.094	25.916	0.680	-1.931*
lnwind	57	25.037	0.529	23.662	25.602	0.999	-2.115**
lnhydro	57	22.414	0.437	20.618	22.827	-1.519	-2.038**
lngeo	57	21.446	0.382	20.500	22.058	0.604	-2.353**
lnbiomass	57	23.256	0.300	22.464	23.763	-0.757	-1.724*
lnmene	53	19.298	0.392	18.421	20.367	-0.737	-2.491**
lnbiofuel	57	22.736	0.737	21.465	23.997	-0.6977	-2.467**

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

4.2. Indirect effect analysis

In Table 5, we explored the economic growth mechanism via which REI influences CO_2E. The findings show that the interaction of solar, wind, and geothermal energy investment and economic growth significantly reduces CO_2E while the interaction of biofuel energy investment and economic growth significantly increases CO_2E. This implies that solar, wind, and geothermal energy investment drives green economic development thereby boosting environmental quality. However, that cannot be said for biofuel energy investment. The interaction findings of solar, wind, and geothermal energy investment and economic growth on CO_2E refute our H_2. However, the interaction findings of biofuel energy investment and economic growth on CO_2E back our H_2.

4.3. Non-linear effect analysis

Table 6 shows the non-linear analysis findings. We observed that the non-linear impact of solar, and wind energy investment on CO_2E is negative and statistically significant while the non-linear impact of small hydropower and biofuel energy investment on CO_2E is positive and statistically significant. This implies that solar and wind energy investment reduce CO_2E only after a threshold is achieved while small hydropower and biofuel energy investment increase CO_2E only after a threshold. Thus, there is an inverted U-shaped relationship between solar and wind energy investment and CO_2E to confirm our H_3 while there is a U-shaped relationship between small hydropower and biofuel energy investment to refute our H_3. The study estimates that the threshold values of solar, and wind energy investment beyond which they reduce CO_2E are 28.125 and 26.308, respectively. Compared to the averages of solar and wind energy investment in Table 2, it is seen that the world operates below the average solar and wind energy investment needed to reduce CO_2E. The 28.125 and 26.308 averages correspond to $1.64 trillion in solar energy investment and $266.3 billion in wind energy investment. This implies that the world needs about $1.64 trillion in solar energy investment and $266.3 billion in wind energy investment on average by 2030 to attain the carbon neutrality target by 2050. Also, the threshold values of small hydropower and biofuel energy investment beyond which they increase CO_2E are 20.071 and 22.8, respectively. These are below the averages of small hydropower and biofuel energy investment as reported in Table 2. This means that the world has surpassed the average small hydropower and biofuel energy investment to contain CO_2E. The 20.071 and 22.8 averages correspond to $520.9 million in small hydropower energy investment and $7.978 billion in biofuel energy investment. Thus, the world should not exceed $520.9 million in small hydropower energy investment and $7.978 billion in biofuel energy investment to attain the carbon neutrality target by 2050.

Aside from the significant results, we found that the non-linear

Table 4
REI and CO_2E.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
lnpop	1.880***	2.539***	2.536***	2.233***	2.374***	2.744***	3.187***
	(0.183)	(0.127)	(0.144)	(0.139)	(0.113)	(0.098)	(0.145)
lngdp	-0.561***	-0.993***	-0.748***	-0.678***	-0.655***	-1.005***	-1.167***
	(0.111)	(0.089)	(0.112)	(0.097)	(0.086)	(0.073)	(0.101)
lntrade	0.451***	0.580***	0.521***	0.537***	0.444***	0.523***	0.513***
	(0.037)	(0.028)	(0.036)	(0.029)	(0.030)	(0.024)	(0.035)
lnsolar	0.021***						
	(0.003)						
lnwind		0.027***					
		(0.004)					
lnhydro			0.016***				
			(0.003)				
lngeo				0.026***			
				(0.003)			
lnbiomass					0.029***		
					(0.003)		
lnmene						0.015***	
						(0.002)	
lnbiofuel							0.015***
							(0.003)
Constant	-22.604***	-34.249***	-35.871***	-29.921***	-33.063***	-38.167***	-46.755***
	(3.218)	(2.138)	(2.334)	(2.316)	(1.838)	(1.604)	(2.442)
Observations	57	57	57	57	57	53	57
R^2	0.993	0.993	0.992	0.994	0.995	0.994	0.992

(...) are standard errors. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 5
Indirect mechanism.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
lnpop	2.412***	2.503***	2.558***	2.328***	2.428***	2.811***	2.756***
	(0.221)	(0.112)	(0.151)	(0.143)	(0.123)	(0.105)	(0.201)
lngdp	2.306***	4.960***	-0.338	1.792	0.511	0.612	-3.847***
	(0.801)	(1.522)	(0.753)	(1.243)	(1.057)	(1.023)	(0.928)
lntrade	0.483***	0.470***	0.519***	0.530***	0.450***	0.543***	0.414***
	(0.034)	(0.038)	(0.036)	(0.028)	(0.030)	(0.027)	(0.047)
lnsolar	1.127***						
	(0.306)						
lnsolar×lngdp	-0.123***						
	(0.034)						
lnwind		2.088***					
		(0.526)					
lnwind×lngdp		-0.229***					
		(0.058)					
lnhydro			0.192				
			(0.320)				
lnhydro×lngdp			-0.019				
			(0.035)				
lngeo				1.097**			
				(0.537)			
lngeo×lngdp				-0.118*			
				(0.059)			
lnbiomas					0.510		
					(0.435)		
lnbiomas×lngdp					-0.052		
					(0.047)		
lnmene						0.810	
						(0.502)	
lnmene×lngdp						-0.087	
						(0.055)	
lnbiofuel							-1.219***
							(0.425)
lnbiofuel×lngdp							0.135***
							(0.047)
Constant	-60.562***	-86.662***	-40.133***	-54.481***	-44.983***	-54.497***	-12.094
	(10.909)	(13.517)	(8.085)	(12.525)	(10.933)	(10.429)	(12.154)
Observations	57	57	57	57	57	53	57
R^2	0.995	0.995	0.992	0.994	0.995	0.995	0.993

(...) are standard errors. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 6
Non-linear analysis.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
lnpop	2.202***	2.594***	2.495***	2.307***	2.372***	2.787***	3.847***
	(0.236)	(0.124)	(0.143)	(0.149)	(0.115)	(0.112)	(0.267)
lngdp	-0.725***	-0.943***	-0.710***	-0.736***	-0.653***	-1.030***	-1.678***
	(0.134)	(0.088)	(0.111)	(0.105)	(0.087)	(0.080)	(0.202)
lntrade	0.487***	0.528***	0.485***	0.541***	0.444***	0.526***	0.602***
	(0.040)	(0.035)	(0.040)	(0.028)	(0.031)	(0.025)	(0.045)
lnsolar	0.225**						
	(0.099)						
$lnsolar^2$	-0.004**						
	(0.002)						
lnwind		0.684**					
		(0.280)					
$lnwind^2$		-0.013**					
		(0.006)					
lnhydro			-0.281*				
			(0.157)				
$lnhydro^2$			0.007*				
			(0.004)				
lngeo				0.375			
				(0.260)			
$lngeo^2$				-0.008			
				(0.006)			
lnbiomass					-0.012		
					(0.338)		
$lnbiomass^2$					0.001		
					(0.007)		
lnmene						-0.100	
						(0.142)	
$lnmene^2$						0.003	
						(0.004)	
lnbiofuel							-0.456***
							(0.164)
$lnbiofuel^2$							0.010***
							(0.004)
Constant	-30.979***	-43.735***	-31.906***	-34.769***	-32.549***	-37.825***	-52.029***
	(5.132)	(4.529)	(3.093)	(4.290)	(4.615)	(1.665)	(2.937)
Observations	57	57	57	57	57	53	57
R^2	0.994	0.994	0.992	0.994	0.995	0.994	0.993
Threshold value	28.125	26.308	20.071	23.438	6.000	16.667	22.8
Threshold amount	$1.64 trillion	$266.3 billion	$520.9 million	$15.1 billion	$403.4 thousand	$17.3 million	$7.978 billion

(...) are standard errors. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 7
FMOLS for REI and CO_2E.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
lnpop	1.878***	2.594***	2.534***	2.279***	2.381***	2.825***	3.204***
	(0.249)	(0.171)	(0.241)	(0.187)	(0.149)	(0.128)	(0.195)
lngdp	-0.591***	-1.081***	-0.730***	-0.729***	-0.660***	-1.076***	-1.216***
	(0.151)	(0.120)	(0.190)	(0.130)	(0.114)	(0.096)	(0.133)
lntrade	0.468***	0.615***	0.532***	0.559***	0.445***	0.526***	0.515***
	(0.049)	(0.038)	(0.059)	(0.038)	(0.039)	(0.032)	(0.047)
lnsolar	0.025***						
	(0.004)						
lnwind		0.034***					
		(0.005)					
lnhydro			0.019***				
			(0.005)				
lngeo				0.029***			
				(0.005)			
lnbiomass					0.033***		
					(0.004)		
lnmene						0.016***	
						(0.003)	
lnbiofuel							0.014***
							(0.004)
Constant	-22.428***	-34.987***	-36.121***	-30.633***	-33.272***	-39.410***	-46.664***
	(4.367)	(2.879)	(3.868)	(3.103)	(2.416)	(2.105)	(3.321)
Observations	56	56	56	56	56	56	56
R^2	0.992	0.992	0.991	0.993	0.994	0.994	0.991

(...) are standard errors. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

impact of geothermal energy investment on CO_2E is negative but not significant while the non-linear impact of biomass and marine energy investment on CO_2E is positive and not significant as well. The discovery also shows that the world needs about $15.1 billion in geothermal energy investment by 2030 to reach the carbon neutrality target while investment in biomass and marine energy investment should not exceed $403.4 thousand and $17.3 million, respectively. These are unique findings to the REI- CO_2E literature.

4.4. Robustness analysis

The FMOLS findings are reported in Tables 7–9. The results of Table 7 show that all the measures of REI have a statistically significant positive impact on CO_2E at the 1% level. Specifically, a 1% increase in solar, wind, small hydropower, geothermal, biomass, marine, and biofuels energy investment increases CO_2E by 0.025%, 0.034%, 0.019%, 0.029%, 0.033%, 0.016%, and 0.014%, respectively. Thus, we can conclude that REI increases global CO_2E. The results also show that population size and trade openness significantly increase CO_2E while economic growth significantly lowers CO_2E to approve our previous findings.

Table 8 shows the FMOLS findings for the economic growth mechanism via which REI influences CO_2E. The findings are similar to those of Table 5, except for the statistically significant negative impact of the interaction effect of marine energy investment and economic growth on CO_2E. This means that marine energy investment indirectly reduces CO_2E by stimulating green economic growth.

Table 9 shows the FMOLS findings for the non-linear impact of REI on CO_2E. Except for the negative statistical impact of the non-linear impact of geothermal energy investment on CO_2E, the discoveries are not different from those reported in Table 6.

4.5. Wavelet analysis

Figs. 4–10 shows the WC analysis for the respective REI measures. The x-axis in both graphs shows time (date), whereas the y-axis denotes frequency. The cone of influence is represented by the grey line. The correlation value is depicted on the right as a vertical line. The dark-blue colour exhibits no proof of association, but the dark-red colour has substantial evidence of correlation. The dark yellow and green colours demonstrate moderate connection. The 5% significance level is indicated by the thick red contour.

In Fig. 4, the majority of arrows are rightward and right down in the short and long run while there is no evidence of correlation in the medium term. Thus, solar energy investment leads to CO_2E in the short and long run. In Fig. 5, the majority of arrows are rightward in both periods, thus wind energy investment leads to CO_2E in all the periods. Fig. 6 shows that the majority of arrows are rightward in the short and long run. Thus, small hydropower energy investment leads to CO_2E in the short and long run. The majority of the arrows are rightward in the short and long run in Fig. 7. This also means that geothermal energy investment leads to CO_2E in the short and long run. In Fig. 8, the majority of arrows tend to the right in the short and medium term. Thus, biomass energy investment leads to CO_2E in the short and medium term. Fig. 9 shows that the majority of arrows tend to the right, right down, and left up suggesting that marine energy investment leads to CO_2E. Finally,

Table 8
FMOLS for the indirect mechanism.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
lnpop	2.471***	2.571***	2.611***	2.362***	2.457***	2.904***	2.749***
	(0.247)	(0.139)	(0.232)	(0.169)	(0.155)	(0.128)	(0.296)
lngdp	2.463**	4.857**	0.001	2.683*	1.052	1.125	-4.340***
	(0.968)	(1.975)	(1.490)	(1.494)	(1.387)	(1.305)	(1.351)
lntrade	0.500***	0.490***	0.530***	0.544***	0.453***	0.554***	0.405***
	(0.038)	(0.047)	(0.057)	(0.033)	(0.039)	(0.034)	(0.065)
lnsolar	1.204***						
	(0.367)						
lnsolar×lngdp	-0.131***						
	(0.041)						
lnwind		2.086***					
		(0.682)					
lnwind×lngdp		-0.228***					
		(0.076)					
lnhydro			0.348				
			(0.622)				
lnhydro×lngdp			-0.036				
			(0.067)				
lngeo				1.495**			
				(0.645)			
lngeo×lngdp				-0.161**			
				(0.071)			
lnbiomas					0.739		
					(0.568)		
lnbiomas×lngdp					-0.077		
					(0.062)		
lnmene						1.096*	
						(0.640)	
lnmene×lngdp						-0.118*	
						(0.070)	
lnbiofuel							-1.413**
							(0.622)
lnbiofuel×lngdp							0.156**
							(0.068)
Constant	-63.427***	-87.443***	-44.489***	-63.434***	-50.709***	-61.357***	-7.374
	(12.771)	(17.383)	(15.006)	(14.995)	(14.212)	(13.281)	(18.041)
Observations	56	56	56	56	56	52	56
R^2	0.994	0.994	0.991	0.994	0.995	0.994	0.992

(...) are standard errors. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 9
FMOLS for non-linear analysis.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
lnpop	2.307***	2.658***	2.460***	2.389***	2.479***	2.827***	4.069***
	(0.273)	(0.152)	(0.238)	(0.182)	(0.137)	(0.152)	(0.338)
lngdp	-0.802***	-1.028***	-0.660***	-0.818***	-0.732***	-1.074***	-1.884***
	(0.155)	(0.110)	(0.189)	(0.129)	(0.104)	(0.109)	(0.258)
lntrade	0.516***	0.552***	0.479***	0.567***	0.480***	0.525***	0.628***
	(0.050)	(0.043)	(0.066)	(0.035)	(0.037)	(0.033)	(0.059)
lnsolar	0.306**						
	(0.121)						
$lnsolar^2$	-0.006**						
	(0.003)						
lnwind		0.681*					
		(0.373)					
$lnwind^2$		-0.013*					
		(0.008)					
lnhydro			-0.467*				
			(0.264)				
$lnhydro^2$			0.011*				
			(0.006)				
lngeo				0.625*			
				(0.318)			
$lngeo^2$				-0.014*			
				(0.007)			
lnbiomass					0.104		
					(0.400)		
$lnbiomass^2$					-0.002		
					(0.009)		
lnmene						-0.015	
						(0.197)	
$lnmene^2$						0.001	
						(0.005)	
lnbiofuel							-0.585***
							(0.213)
$lnbiofuel^2$							0.013***
							(0.005)
Constant	-33.766***	-44.520***	-29.541***	-38.680***	-35.772***	-39.140***	-53.808***
	(5.971)	(5.807)	(5.257)	(5.256)	(5.432)	(2.248)	(3.800)
Observations	56	56	56	56	56	52	56
R^2	0.993	0.993	0.991	0.994	0.994	0.994	0.992

(...) are standard errors. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

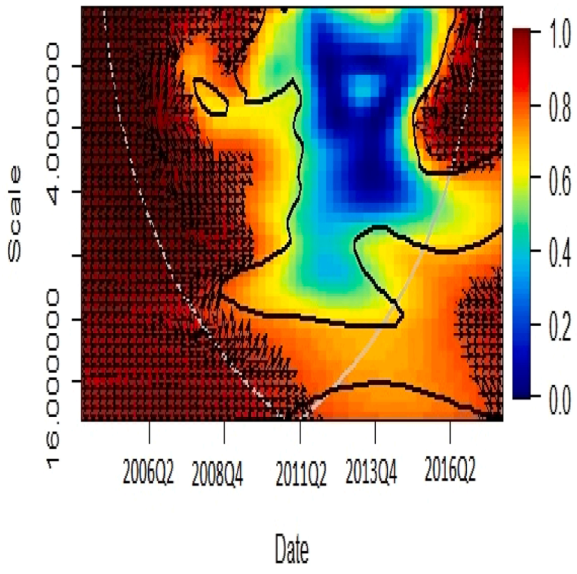


Fig. 4. WC analysis for solar energy investment.

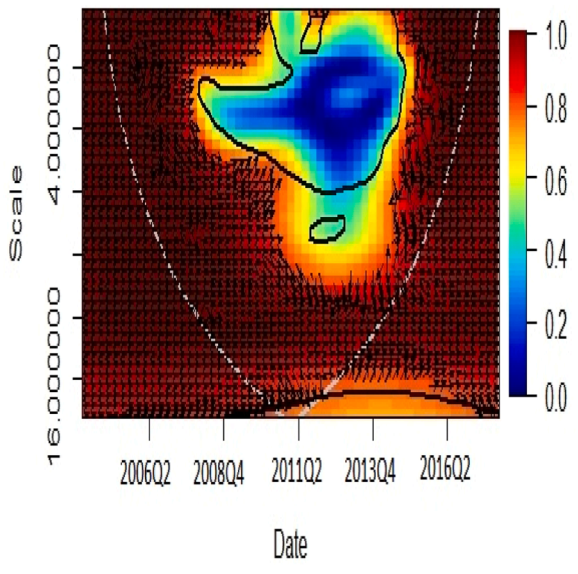


Fig. 5. WC analysis for wind energy investment.

Fig. 10 shows that the majority of arrows are rightward and right down in the short and medium term while in the long term, the arrows tend left and left down. Thus, biofuel energy investment leads to CO_2E in the short and medium term while CO_2E leads to biofuel energy investment in the long run.

5. Conclusion

Rapid REI is necessary to help the world reach its carbon neutrality

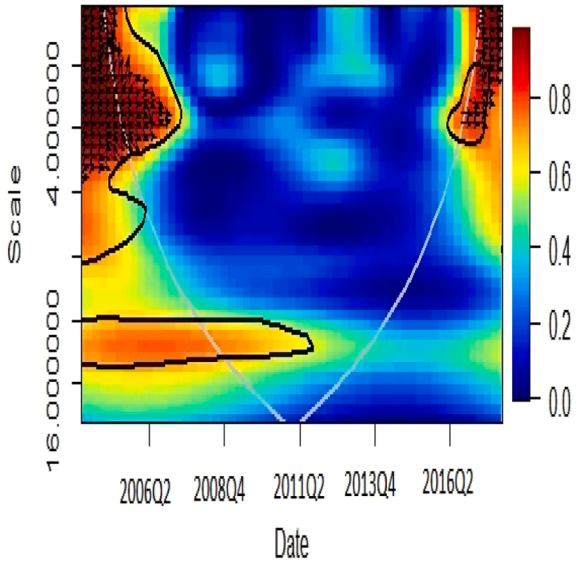


Fig. 6. WC analysis for small hydropower energy investment.

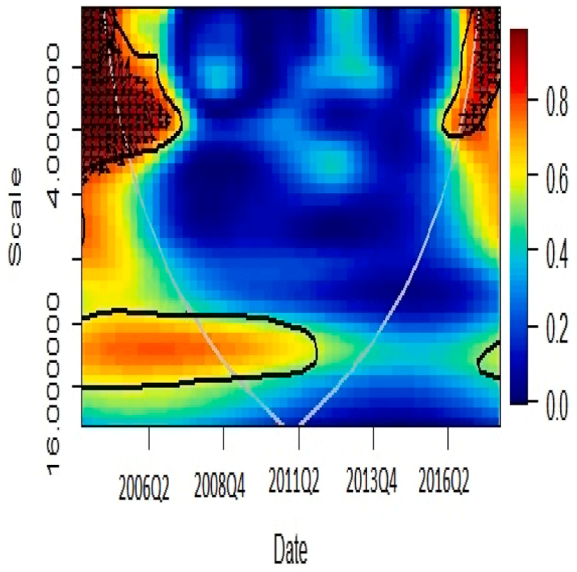


Fig. 7. WC analysis for geothermal energy investment.

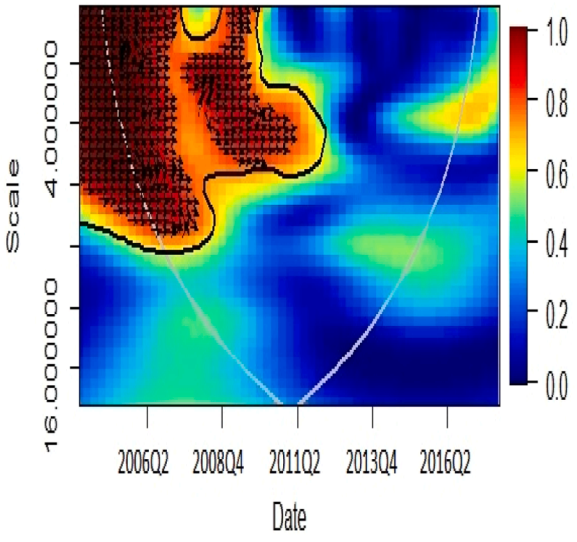


Fig. 8. WC analysis for biomass energy investment.

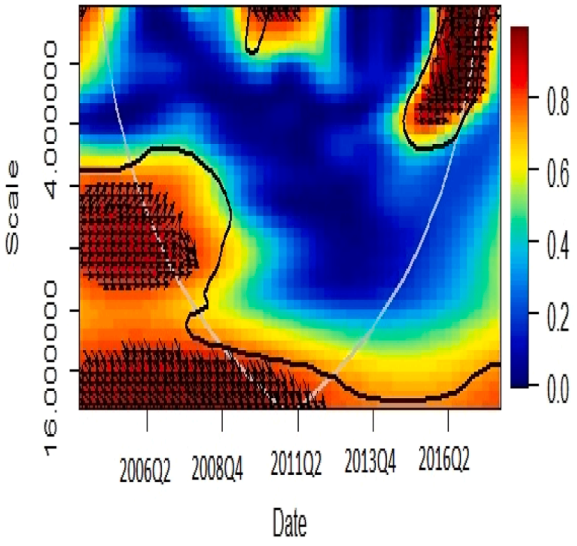


Fig. 9. WC analysis for marine energy investment.

target by 2050; however, whether REI impacts CO_2E remains conflicting. This study, therefore, examined the linear, non-linear, economic growth mechanism via which REI influences CO_2E and the interdependence between REI and CO_2E using a global dataset. While using the OLS estimator and the WC analysis, we found that all the measures of REI significantly increase CO_2E from the linear analysis. However, the non-linear analysis showed that solar and wind energy investment have a statistically significant inverted U-shaped relationship with CO_2E while small hydropower and biofuel energy investment have a statistically significant U-shaped relationship with CO_2E. Thus, solar and wind energy investments reduce CO_2E only after attaining a certain threshold of investment. The threshold values of solar and wind energy investment are 28.125 (representing about $1.64 trillion) and 26.308 (representing about $266.3 billion), respectively. Also, the threshold values of small hydropower and biofuel energy investment beyond which they increase CO_2E are 20.071 (representing about $520.9 million) and 22.8 (representing about $7.978 billion), respectively. Furthermore, solar, wind and geothermal energy investment indirectly reduce CO_2E by increasing green economic growth while biofuel energy investment indirectly increases CO_2E by increasing economic growth. Finally, all the measures of REI lead to CO_2E in both the short and long run, except biomass and biofuel energy investment.

The findings of this study call for further investment in wind and solar energy projects to attain the carbon neutrality target. Accordingly, we recommend production tax credits and investment tax credits. These tax credits can be used as substitutes for investing in these technologies and can result in reductions in income or corporate taxes. Feed-in tariffs can also be established to encourage the deployment of solar and wind energy technologies. Policymakers can also provide subsidies to solar and wind energy companies and redirect a portion of fossil fuel subsidies toward solar and wind energy. Additionally, implementing RE education at all levels can help enlighten citizens about the benefits of RE and encourage mass adoption of these technologies. Special financial tools such as carbon pricing, affordable financing, and supporting research and development can also be adopted to encourage the consumption of solar and wind energy technologies. The findings of a positive effect of population size on CO_2E imply that the world needs to implement rapid measures to reduce the environmental impact of the increasing

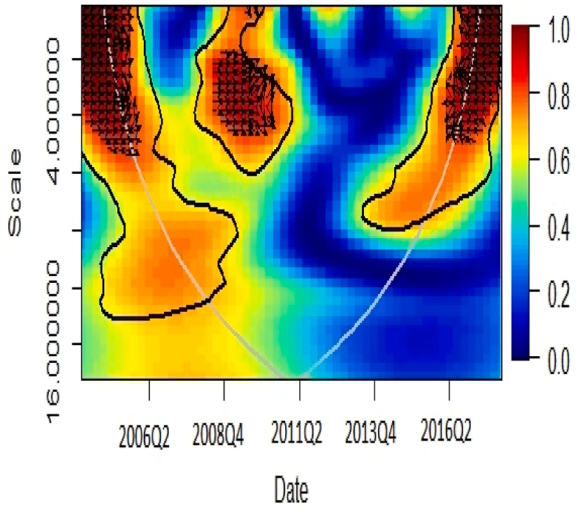


Fig. 10. WC analysis for biofuels energy investment.

population. Policymakers can promote the adoption and use of public transportation systems like buses, trains, subways, and bikes to help lower the environmental impact of rapid population growth. The positive impact of trade openness on CO_2E also implies that cleaner production and distribution technologies should be adopted to contain environmental pollution emanating from trade. Policymakers can also encourage the production and transportation of goods using RE sources like solar and wind energy.

While this study primarily focused on global datasets, there are specific REI targets for countries and regions. Therefore, future research can focus on country or region-specific analyses. Furthermore, due to multicollinearity challenges and our small number of observations, we studied the impact of economic growth, population, trade openness, and REI on CO_2E. However, several other factors such as energy efficiency influence CO_2E. Accordingly, future studies can test the robustness of our findings by including additional control variables such as energy efficiency in the models when more data becomes accessible in the future. Finally, this study was restricted to 2018 because of data-related obstacles. As a result, forthcoming investigations should use the most up-to-date dataset to revisit the relationship between REI and CO_2E when new data becomes available.

CRediT authorship contribution statement

Isaac Appiah-Otoo: Conceptualization, Writing – review & editing, Methodology, Software, Data curation, Investigation, Visualization. **Xudong Chen:** Conceptualization, Writing – review & editing, Methodology, Software, Data curation, Investigation, Visualization, Supervision. **Matthew Binyam Kursah:** Conceptualization, Writing – review & editing, Data curation, Supervision. We, the authors, acknowledge that this manuscript has not been submitted to a different journal, and there is no conflict of interest.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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